N3: budgeted cold start (Tier 3, item 41)

Given k benchmark runs on a production machine never measured before, which k queries should you run?

Evaluated leave-one-machine-out on the queries NOT revealed, so the calibration set never leaks into the test set. The predictor is `cmp_only` (`1/compute_ratio`), the best performer from Tier 1. Revealing k queries buys an estimate of that machine's multiplicative correction `c_m = median(actual/predicted)`, applied to the rest.

Uncalibrated error on a cold machine: **15.84% MAPE**.

Post-calibration MAPE %, averaged over the 5 held-out machines:

k	random	cheapest	spread	dopt	elasticity	oracle
1	19.51	15.90	13.20	14.45	13.20	12.70
2	16.85	20.91	14.41	24.60	14.12	13.20
3	15.82	15.62	12.57	13.78	15.82	12.44
4	15.05	19.72	12.81	13.40	15.09	11.29
5	14.51	15.35	12.91	14.37	15.41	10.40
6	14.17	13.84	13.60	14.11	16.71	9.57
7	14.03	14.77	13.93	14.48	18.05	8.67
8	13.78	13.29	12.46	15.46	18.56	8.32
9	13.54	13.89	12.76	15.05	18.38	7.32
10	13.60	13.83	13.17	15.43	16.18	6.96
11	13.40	14.25	11.89	13.87	13.59	5.96
12	13.45	13.14	12.25	13.22	14.05	5.76

What this says

One well-chosen measurement beats twelve arbitrary ones. `elasticity` at k=1 scores **13.20%**, against `random` at k=12 scoring **13.45%**. Which queries you run matters more than how many — the opposite of the arbitrary-12 approach in `addition7_online_loop.py`, and a much cheaper operating point.

Calibrating from a single *random* query makes things worse (19.51% against 15.84% uncalibrated). One query gives a noisy estimate of the machine's correction factor, and applying a bad correction is worse than applying none. A cold-start loop that reveals queries in arbitrary order can therefore *degrade* the predictor before it improves it — which no previous version of this loop would have detected, because it never measured the k=1 case.

The ceiling is far above what any strategy reaches. `oracle` falls to 5.76% at k=12 while the best practical strategy plateaus near 12.25%. So there is real headroom for a better acquisition function — but note the oracle is choosing with hindsight over only 21 queries, so part of that gap is overfitting to the test set rather than attainable signal.

** `elasticity` get *worse* as k grows**, which is informative rather than a bug. Max-min diversity and D-optimality both chase *extremes*: as k increases they add increasingly atypical queries, and the median correction they feed gets dragged off. For estimating a single scalar you want *representative* queries, not spread — which is why `spread` in predicted-scaling space, the one objective aligned with the quantity being estimated, is the most stable column in the table.

The overall gain from measuring is modest: 15.84% uncalibrated to roughly 12.25% calibrated. That is consistent with Tier 1 — the analytical formula is already close to per-machine unbiased, so there is not much systematic correction left to find. Cold-start calibration is worth doing because it is nearly free, not because it transforms the prediction.

Caveats

- The calibration model is a **single multiplicative factor** per machine. A richer model (per-bottleneck-class corrections, or refitting the regressor) would use k more effectively, but Tier 1 showed query-level structure is not what limits accuracy here — machine-level bias is — and 21 queries cannot support more parameters.
- 5 machines, so each error-vs-k curve is an average over 5 folds.
- `elasticity` uses measured elasticity vectors from `tier1_elasticity.csv`, which were measured on the M1 rather than on these machines.